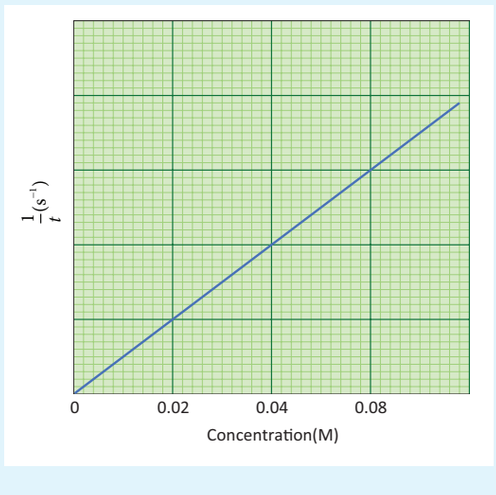
Activity

1. Take three conical flasks and label them as A, B, and C.
2. using a burette, add 10, 20 and 40 ml of 0.1M sodium thiosulphate solution to the flask A, B and C respectively. And then add 40, 30 and 10 ml of distilled water to the respective flasks so that the volume of solution in each flask is 50ml.
3. Add 10 ml of 1M HCl to the conical flask A. Start the stop watch when half of the HCl has been added. Shake the contents carefully and place it on the tile with a cross mark as shown in the figure. Observe the conical flask from top and stop the stops watch when the cross mark just becomes invisible. Note the time.
4. Repeat the experiment with the contents on B and C. Record the observation.

Flask	Volume of $Na_2S_2O_3$	Volume of water	Strength of $Na_2S_2O_3$	Time taken (t)
A	10	40	0.02	
B	20	30	0.04	
C	40	10	0.08	

Draw a graph between $\frac{1}{t}$ Vs concentration of sodium thiosulphate. A graph like the following one is obtained.



$\frac{1}{t}$ is a direct measure of rate of reaction and therefore, the increase in the concentration of reactants i.e $Na_2S_2O_3$, increases the rate.



7.9.3 Effect of surface area of the reactant:

In heterogeneous reactions, the surface areas of the solid reactants play an important role in deciding the rate. For a given mass of a reactant, when the particle size decreases surface area increases. Increase in surface area of reactant leads to more collisions per litre per second, and hence the rate of reaction is increased.For example, powdered calcium carbonate reacts much faster with dilute HCl than with the same mass of $CaCO_3$ as marble

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